# Sleep Hygiene Guidance

Reference notes for a wearable-data recovery assistant. Adapted and paraphrased from general public-health sleep guidance (CDC, Harvard Health, sleep medicine consensus statements). Not medical advice.

## Why sleep matters for recovery

Sleep is the primary window in which the body repairs muscle tissue, consolidates memory, regulates hormones, and restores the nervous system after training stress. Both the total amount of sleep and its quality (how continuous and undisturbed it is) affect next-day recovery, not just duration alone.

Consistently short or fragmented sleep is associated with slower physical recovery, reduced glucose regulation, elevated resting heart rate, lower next-day heart rate variability, and a higher perceived effort during exercise.

## How much sleep is typically recommended

General population guidance suggests most healthy adults need about 7-9 hours of sleep per night for full physiological recovery. Athletes or people with high training loads often benefit from the higher end of that range, or slightly more.

Sleep needs vary by individual, age, and current training load, so a personal baseline built from a person's own recent history is usually more useful than a single fixed number.

## Core sleep hygiene habits

- Keep a consistent sleep and wake time, including on weekends, so the body's internal clock stays stable.
- Keep the bedroom quiet, dark, and cool, and reserve it mainly for sleep.
- Turn off or dim bright screens for roughly 30-60 minutes before bed.
- Avoid large meals, alcohol, and caffeine in the hours before bedtime; caffeine sensitivity means some people need to stop even earlier in the day.
- Get daytime physical activity and natural light exposure, which reinforce the sleep-wake cycle.
- Keep naps short (30 minutes or less) and earlier in the day if they are needed.

## Reading a sleep log for a recovery assistant

When comparing a single night to a person's baseline, a modest one-night shortfall (for example, an hour less than usual) is normal and usually only calls for a small adjustment in training intensity.

A pattern of multiple consecutive nights of short or poor-quality sleep is a stronger signal and should weigh more heavily on a recovery score than one bad night in isolation.

Frequent, unexplained trouble falling asleep or staying asleep, especially alongside daytime fatigue, is a cue to suggest the person speak with a healthcare provider rather than just adjusting a workout.

## Guidance for the assistant's tone

Sleep advice from this assistant should stay practical and behavioral (routine, environment, timing) and should never attempt to diagnose a sleep disorder. Recurring or severe sleep problems should always be redirected to a clinician.